(*) Some pins in ESP 23 are capacitance (touch) pins

CAPACITANCE CALC

```
from machine import Pin, TouchPad
import time
touch = TouchPad(Pin(12))
while True:
    touch_value = touch.read()
    print(touch_value)
    time.sleep(0.4)
```

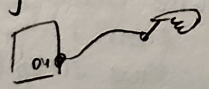
← setup touchpad

↳ Pin(12, Pin.IN) ↓ default ∴ optional.

← provides value in 0.4 seconds interval.

(*) (↑) Capacitance (400<) if you don't touch other end of jumper wire.

(↓) Capacitance (<100) if you touch



LIGHTING LED when capacitance <100 (using touch)

```
from machine import Pin, TouchPad
import time
touch = TouchPad(Pin(4))
l = Pin(2, Pin.OUT)
while True:
    touch_value = touch.read()
    print(touch_value)
    time.sleep(0.4)
    if touch_value <100:
        l.on()
        time.sleep(1)
        l.off()
        time.sleep(1)
```

} import modules

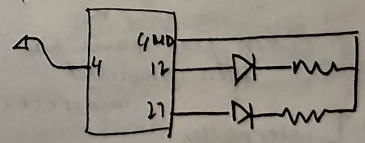
} define variables for light 1 capacitance indicator.

For (EXTERNAL)

l = Pin(12, Pin.OUT)

↓ Anything other than 2

2 LEDs, Touch pin, resistors, ESP 32



Code ⇒

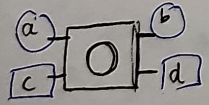
```
from machine import Pin, TouchPad
import time

t = TouchPad(Pin(4))
l1 = Pin(12, Pin.OUT)
l2 = Pin(27, Pin.OUT)
while True:
    tv = t.read()
    time.sleep(0.4)
    if tv<300:
        print("Pin 4 is touched")
        l1.on()
        time.sleep(1)
        l1.off()
    else:
        print("Not touched")
        l2.on()
        time.sleep(1)
        l2.off()
print(tv)
```

ERRORS | DEBUGGING

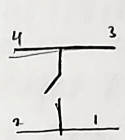
- from machine → Import Error
- Indentation Error
- "module not found"
- Syntax Error.
- Attribute error. (when func^n name has been provided incorrect) "Pin"

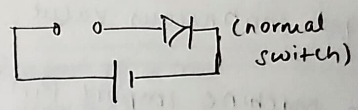
PUSH BUTTON

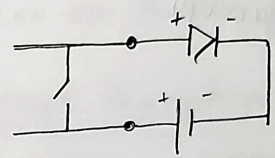


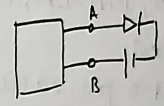
a & b, c & d] internally connected

When you press button, top & bottom get connected to eachother.

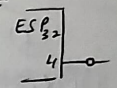


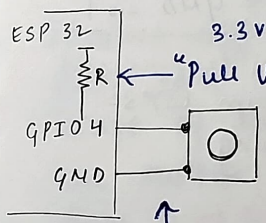






If a pin isn't connected ⇒ FLOATING cond^n





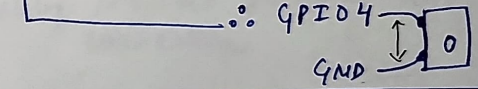
3.3 V

"Pull Up" Resistor

(GND) Ground ⇒ logic 0

GPIO ⇒ logic 1

∴ GPIO 4



GPIO4 = GND = logic 0

To see VALUE of Push Button

```
from machine import Pin
import time
pb = Pin(4, Pin.IN, Pin.PULL_UP)
pb_val = pb.value()
print(pb_val)
```

OUTPUT

1

To get continuous value

```
from machine import Pin
import time
pb = Pin(4, Pin.IN, Pin.PULL_UP)
while True:
    pb_val = pb.value()
    print(pb_val)
```

(+) time.sleep(0.2)

By pushing button, GPIO 4 = GND = 0

(*) We used 2 jumper wires & isolated the push button away from ESP 32 coz pins were overlapping.

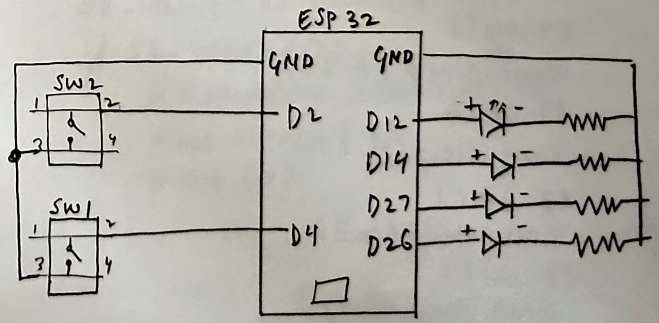
ACTIVITY

Right & Left Indicator of a car
2 Push Button + 4 LEDs
When no Button pressed = 4 LEDs ON.

Concept Description

Components : 4 LEDs.
2 Push Buttons.
ESP 32

Involves different time durations for each LED.
Usage of loops (while, if, else)
One switch activates series of blinking from R → L and the other, L → R.
Probably involves parallel LED connections



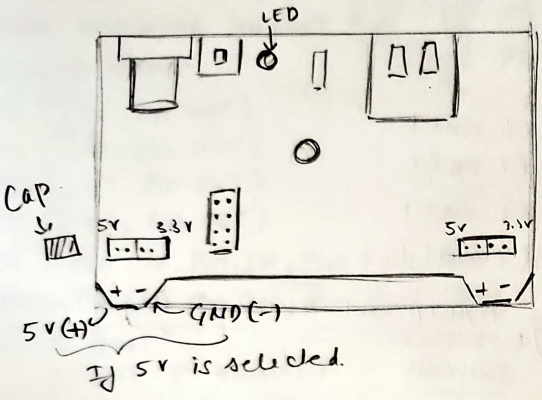
CODE

```
from machine import Pin
import time
l1 = Pin(12, Pin.OUT)
l2 = Pin(14, Pin.OUT)
l3 = Pin(27, Pin.OUT)
l4 = Pin(26, Pin.OUT)
p1 = Pin(18, Pin.IN, Pin PULL_UP)
p2 = Pin(19, Pin IN, Pin. PULL_UP)
while True:
    p1_val = p1.value()
    if p1_val == 0:
        l1.off()
        time.sleep(0.2)
        l2.off()
        time.sleep(0.22)
        l3.off()
        time.sleep(0.24)
        l4.off()
        time.sleep(0.26)
        l4.on()
        time.sleep(0.28)
        l3.on()
        time.sleep(0.3)
        l2.on()
        time.sleep(0.32)
        l1.on()
        time.sleep(0.34)
        l1.off()
        l2.off()
        l3.off()
        l4.off()
    else:
        l1.on()
        l2.on()
        l3.on()
        l4.on()
```

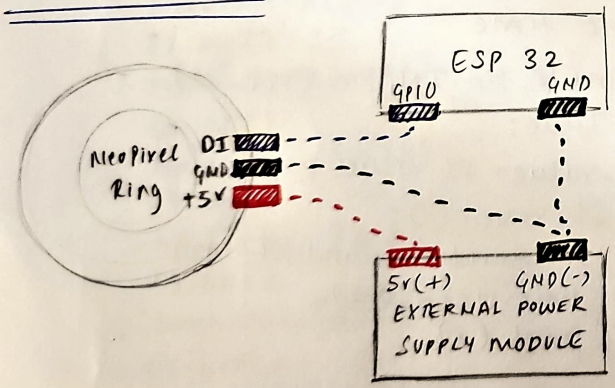
Generating RANDOM number by pushing switch

```
from machine import Pin
import random
import time
S1 = Pin(18, Pin.IN, Pin.PULL_UP)
while True:
    S1_value = S1.value()
    if S1_value == 0:
        r = random.randint(1,10)
        time.sleep(0.2)
        print(r)
```

POWER MODULE



NEO PIXEL RING




```
from machine import Pin
import time
import neopixel
np = neopixel.NeoPixel(Pin(4), 16)
```

No. of LEDs.

```
# name[INDEX] = (R,G,B)
# For red (255,0,0), For Green (0,255,0)
  For blue (0,0,255)
np[1] = (255,0,0)
np[2] = (0,255,0)
np[3] = (0,0,255)
np.write()
```

define np[0] → np[15] for all LEDs to light up.

3/2/26

GENERATION of random no. every second

```
import random
import time
while True:
    r = random.randint(1,10)
    time.sleep(1)
    print(r)
```

MANUAL looping and linking of neopixel

```
from machine import Pin
import time
import neopixel
np = neopixel.NeoPixel(Pin(12),16)
```